MixerReturn user guide

MixerReturn **bolts a Dugan Automixer onto a console that hasn't got one** — post-fader, without spending an insert slot or a second channel strip per mic.

It is a VST3/AU/Standalone plugin that gives a plugin host something it does not otherwise have: **a shared summing bus spanning many instances**. Put one instance after the automixer on each channel's rack, and one more instance set to output the sum. That sum returns to the desk as a mix's External Input.

> **Before you rely on this:** the summing bus is verified numerically against the actual shipped processor class — a one-block delay with the summing instance processed both first and last, 24 senders with the host's processing order reshuffled every block, 400 blocks with 17 instances racing on their own threads — and it is clean under ThreadSanitizer. `pluginval` passes at strictness 8.
>
> It **has** been loaded in Waves SuperRack Performer v15.15.12 on macOS: two instances in separate racks both report *"2 members on this bus"*, which confirms in the real host that instances share one registry. **No audio device was attached for that test**, so audio has not been heard passing through the sum. It has **not been run against a real console and not been used on a show**. **Every claim about SQ behaviour here comes from the reference guide, not from hardware.** Review it before use on live gear.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

Why this exists

SuperRack's patcher allows **only one rack per output I/O**, so twenty-four rack outputs cannot be patched onto one output to sum them. The desk can't sum them either without spending twenty-four input channels to do it.

MixerReturn fills that one gap. Everything else about the design follows from it.

What it doesn't cost you

Because the automixer works on the **direct outs** rather than on channel inserts:

- **The insert slots stay free.** You can still insert plugins on the channel strips exactly as normal — the automixer isn't competing for them.

- **The automixer sits post-fader**, without relocating any insert points.
- **No channel is used twice.** One channel strip per mic, as usual.

It adds an automixer to a desk without taking anything away from it.

The signal flow

```
SQ channels 1-24
  |  direct outs, set to follow the fader
  v
SLink / Dante / Waves card  ---->  SuperRack Performer
                                     Rack 1..24:  Dugan Automixer → MixerReturn (Send)
                                     Rack 25:                      MixerReturn (Bus Sum)
                                                                          |
SQ mix "External Input"  <-------------------------------------------------
```

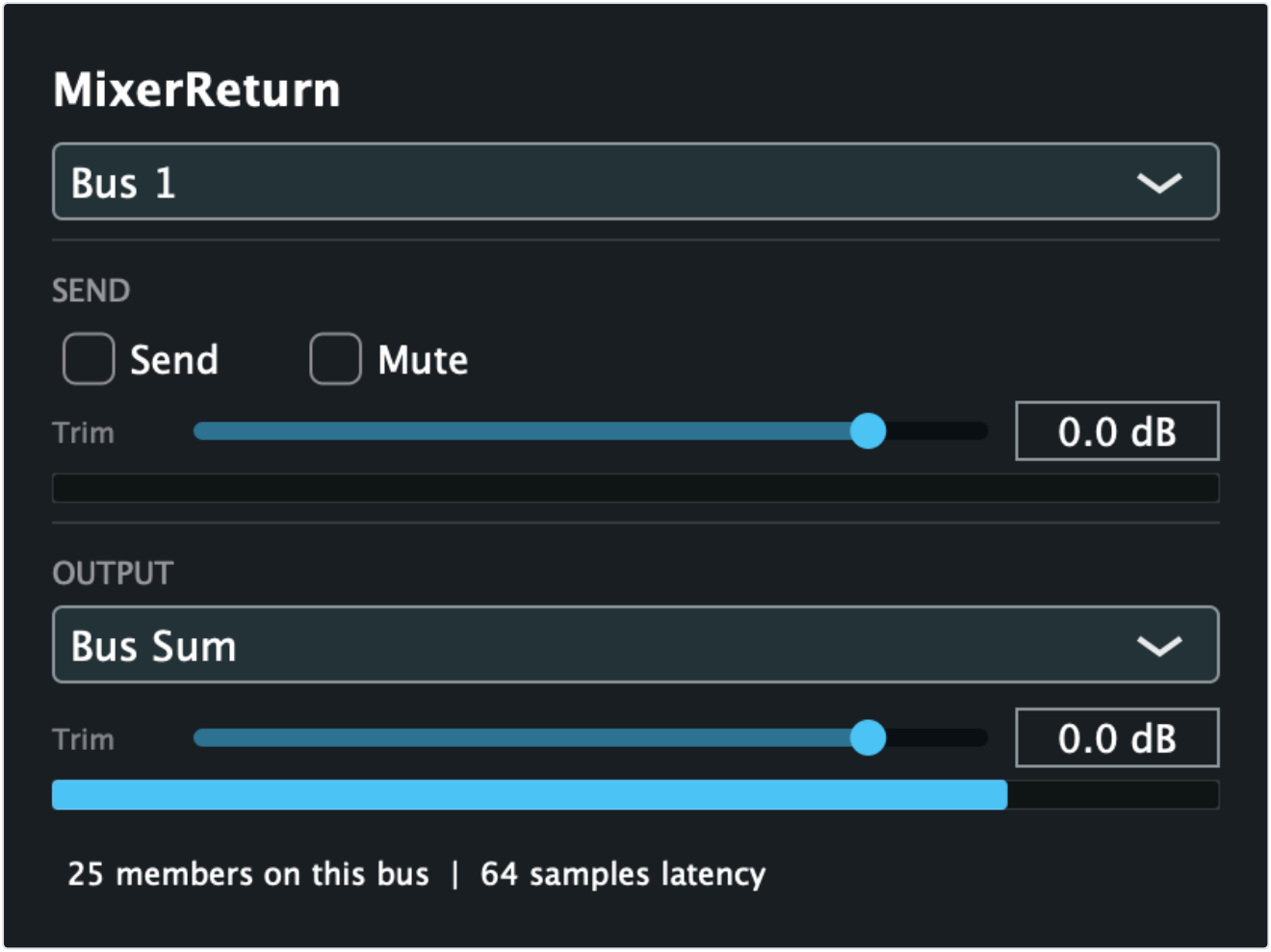
The operator keeps working ordinary channel strips. Dugan sees post-fader signals.

The two roles

Sending and receiving are independent, so an instance can be a sender, a receiver, or both.

On each channel, after the automixer: **Send** enabled, **Output** set to `Input` so the rack passes its own audio through.

On the return rack: **Send** disabled — so the send meter is empty — and **Output** set to `Bus Sum`.



Both rendered from the real editor with a full 24-channel rig registered on the bus, which is why the member count and latency readout are actual values rather than a mock-up.

The **member count** is the readout to trust when setting up: if it does not say what you expect, an instance is on the wrong bus or hasn't loaded.

The controls

Control	What it does
Bus	Which of the 8 shared buses this instance joins.
Send / **Mute**	Whether this instance contributes its input to the bus.
Send Trim	−inf to +10 dB on the contribution, matching the SQ's own direct-out trim range.
Output	`Input` (pass through), `Bus Sum` (emit the sum), or `Input + Bus Sum`.
Output Trim	−inf to +10 dB on this instance's output.

The one thing to know

The bus sum is delayed by exactly one block, identically for every sender, no matter what order the host processes the instances in.

That uniformity is deliberate, and it is the reason the design looks the way it does: a shorter but *uneven* delay across channels would comb-filter the sum. See DESIGN.md.

Setting it up on an Allen & Heath SQ

Two console-side details that will bite otherwise.

> **"Follow Fader" on direct outs is global across all 48 channels.** There is no post-fader tap point; post-fader comes from that one switch, and it is all-or-nothing.
>
> **If you also multitrack off direct outs, that recording becomes post-fader too.** Route multitrack over **tie lines** — post-preamp, bypassing the processing path — instead.

> **Unroute the automixed channels from the main mix.** The summed return must be their only path there. A channel reaching the mix both directly *and* through the return will comb.
>
> Pre-fader monitor sends stay inside the desk and are fine.

The SQ core runs at 96 kHz, so the plugin host must too.

Troubleshooting

Symptom	Cause
Member count is lower than expected	An instance is on a different bus, or hasn't loaded. The count is the authoritative readout.
Comb filtering on the automixed channels	They are reaching the main mix both directly and through the return. Unroute them from the mix.
Automixer is reacting to the wrong level	Direct outs are pre-fader. "Follow Fader" is one global switch on the SQ.
Multitrack recording levels now follow the faders	Same switch — it is global. Move multitrack to tie lines.
Bus Sum instance outputs nothing	Check **Output** is `Bus Sum` and not `Input`, and that senders are on the same bus.
Send meter is empty on a channel instance	**Send** is disabled, or **Mute** is on.
Sample-rate mismatch with the console	The SQ core is 96 kHz; the host must match.

See also

- DESIGN.md — the summing bus, and why the delay is uniform by design
- README — signal flow, controls and downloads

Installing

On macOS, install to the system folder, not your home folder:

```
/Library/Audio/Plug-Ins/VST3/
```

This matters more than it looks. **Waves SuperRack Performer scans only the system VST3 folder — it never looks in `~/Library/Audio/Plug-Ins/VST3/`.** A plugin dropped into the user folder simply will not appear in SuperRack, with no error and nothing in a log to explain it. The `.pkg` installer puts it in the right place; if you use the `.zip`, copy the bundle to the path above rather than to your home folder.

That directory is world-writable on a stock macOS install, so no administrator password is needed to copy into it.

SuperRack scans at launch only. If you install or update the plugin while SuperRack is running, quit and relaunch it before expecting to see the change.

You do not need to unquarantine anything. A locally built or `.zip`-extracted bundle carries no `com.apple.quarantine` attribute here, and the plugin loads and runs in SuperRack with an ad-hoc signature — Developer ID signing and notarization are not required for it to be hosted.

Where this is going

The device is the product; the plugin is the version that works without installing one.

Once the wrapper exists, a rack's output patch *is* the routing decision — physical output to behave as an insert, or a `Sum` port to feed the buses — with no plugin in the chain at all. The wrapper is also strictly better technically: a driver receives every Sum port in one callback, so it can sum with **no added delay**, where the plugin must cost a block. The entire two-page barrier in this plugin exists only because plugin instances cannot see each other's timing; a driver has no such problem.

What the plugin still gives you, honestly and completely: it exists today; it needs no driver install, no admin rights, no notarized system extension and no signed kernel driver; it does not depend on successfully wrapping any particular interface; and it therefore runs on locked down or rented machines where a system audio device is not an option. That is "easier to deploy", not "does something the device cannot". The one capability gap runs the other way — routing between separate applications, which a per-process plugin bus can never do.
